CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 4: Minimum Spanning Tree with Prim & Kruskal Algorithms

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #4 | Academic Year: 2026-27

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1. Overview & Objective:

Graph algorithms implementation comparing Disjoint Set Union and Priority Queue Min-Heap.

2. Practical Tasks & Verification Steps:

1. Represent weighted undirected graph using adjacency list and edge list structures.
2. Implement Kruskal algorithm sorting edges by weight and detecting cycles using Union-Find with path compression.
3. Implement Prim algorithm starting from arbitrary vertex expanding MST using PriorityQueue.
4. Output total minimum spanning tree weight and set of selected spanning tree edges.

Submission Guidelines & Verification:

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